

# Jonas Lim Hong Xiang

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Senior ML engineer with 7 years delivering production AI/ML systems end-to-end: from problem framing and feature engineering through model deployment, monitoring, and iteration. Strong depth across classical ML, deep learning, and applied GenAI – including agentic AI systems with tool use, multi-step execution, and reliability patterns. Experienced shipping decisioning and anomaly detection models at scale with measurable business impact.

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## PROFESSIONAL EXPERIENCE

### Co-Founder & Software Architect

May 2024 - Jun 2026

NockHub, Singapore | [nockhub.com](https://nockhub.com)

- Owned end-to-end delivery of production Python services and agentic AI workflows, including tool-calling, automated decision steps, and reliability patterns (retries, fallbacks, idempotent operations).
- Evaluated and integrated GenAI/LLM tooling for automated pipelines; established model selection criteria based on quality, latency, cost, and security requirements.
- Defined engineering and delivery standards across a team of 3 engineers; maintained CI/CD, deployment, and observability practices across all production workstreams.
- Collaborated with non-technical stakeholders to translate business requirements into model metrics and system specifications.
- Grew platform to 4 ranges, 7,000+ users, and 4,000+ bookings per month.

### Senior Machine Learning Engineer

Jun 2022 - Jul 2024

Carro, Singapore

- Owned end-to-end delivery of multiple production ML systems: predictive pricing (regression/ranking), vehicle damage detection (classification/anomaly detection), and engine acoustics (time series/sequence modeling).
- Built scalable Python ML microservices on GPU clusters serving 12,000+ requests per day; optimized for reliability, low latency, and cost efficiency.
- Led model selection and production transition from Mask R-CNN to YOLOv5 based on quality, latency, and cost trade-off analysis; reduced inference time from 500ms to 50ms with no production disruption.
- Built feature engineering pipelines, a feature store, and a metadata store; established offline evaluation datasets and quality gates that improved training data quality by 50% and enabled reproducible experiments.
- Implemented automated training, deployment, and monitoring pipelines; introduced drift detection and error analysis that reduced operational overhead by 75 engineer-hours per month.
- Replaced monolithic data crawler with distributed architecture; reduced processing time from 7 days to 9 hours and improved data quality and lineage tracking.
- Partnered with product, operations, and data teams across Singapore, Malaysia, Indonesia, and Taiwan; translated business risk objectives into model metrics and communicated trade-offs to technical and non-technical stakeholders.

### Data Scientist

Jun 2019 - May 2022

Micron Technology, Singapore

- Built an anomaly detection and classification model within semiconductor quality controls; improved inspection tool capacity by 10%.
- Built and deployed a predictive decisioning model on-premises within compliance constraints, contributing to USD 11M in cost savings.
- Optimized SQL data pipelines for model training; reduced query time by 50% and improved data reliability for downstream ML workflows.
- Delivered end-to-end ML solutions from problem framing to production deployment, collaborating across engineering, quality, and operations teams.

EDUCATION & AWARDS

Master of Computer Science (Data Science & AI)

Aug 2024 - Jun 2026

University of Sydney, Australia

Distinction | Singapore IMDA SG Digital Scholar

- Capstone: Designed and deployed an Agentic AI system on a Unitree Go2 quadruped robot – built LLM-driven agent with tool use, task decomposition, multi-step execution, and reliability patterns for real-time autonomous interaction.

Bachelor of Engineering (Aerospace)

Aug 2015 - May 2019

Nanyang Technological University, Singapore

First Class Honors, Highest Distinction

TECHNICAL EXPERTISE

**Agentic AI & LLM:** tool use, task decomposition, multi-step execution, RAG, prompt engineering

**ML Systems:** feature pipelines, model evaluation, drift detection, automated deployment

**Languages:** Python, SQL, TypeScript, Bash

**Cloud & Infrastructure:** AWS, GCP, Terraform

**ML/AI:** PyTorch, TensorFlow, classification, regression, anomaly detection, time series, embeddings

**Data Engineering:** PySpark, SQL, ETL design, data quality & lineage

**Backend & APIs:** FastAPI, Docker, distributed systems

**Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, RDS

HACKATHONS

2025 Global Health Systems Hackathon Hub, "Building High-Value Health Systems: Leveraging AI". Team I Know You Know won 2nd place in Sydney and placed Top 20 globally.

CERTIFICATIONS

AWS Certified Solutions Architect - Professional

Jan 2024

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